

In-Class Lab: Mean-Based Z-Test Questions (revised)

Instructions

For each question, follow these steps:

1. **State the Null Hypothesis (H_0) and Alternative Hypothesis (H_1).**
 2. Calculate the Z-score and p-value.
 3. Compare the p-value with the significance level ($\alpha = 0.05$).
 4. Draw a conclusion: Decide whether to reject or fail to reject the null hypothesis.
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Question 1: Average Time on a Website

Scenario: A website owner claims that the average time a visitor spends on their site is 10 minutes. A sample of 100 visitors shows an average time of 9.8 minutes with a population standard deviation of 1.5 minutes.

Task: Test the website owner's claim at a 5% significance level.

Question 2: Battery Life of Smartphones

Scenario: A smartphone manufacturer claims that the average battery life of their new model is **at least** 20 hours. A sample of 80 phones shows an average battery life of 19.5 hours with a population standard deviation of 1.2 hours.

Task: Test the manufacturer's claim at a 5% significance level.

Question 3: Average Weekly Hours Worked

Scenario: A company claims that their employees work an average of 40 hours per week. A sample of 150 employees shows an average workweek of 39.6 hours with a population standard deviation of 2.0 hours.

Task: Test the company's claim at a 5% significance level.

Question 4: Weight of Packaged Goods

Scenario: A food processing company claims that their packaged goods weigh an average of 500 grams. A sample of 120 packages shows an average weight of 498 grams with a population standard deviation of 5 grams.

Task: Test the company's claim at a 5% significance level.

Question 5: Average Speed of Delivery

Scenario: A delivery company claims that the average delivery time for their services is **less than** 3 hours. A sample of 60 deliveries shows an average delivery time of 3.2 hours with a population standard deviation of 0.6 hours.

Task: Test the company's claim at a 5% significance level.

Question 6: Average Daily Water Consumption

Scenario: A city claims that the average daily water consumption per household is **more than** 160 liters. A sample of 90 households shows an average consumption of 158 liters with a population standard deviation of 10 liters.

Task: Test the city's claim at a 5% significance level.

Question 7: Average SAT Scores

Scenario: A school district claims that their students have an average SAT score of 1100. A sample of 200 students shows an average score of 1085 with a population standard deviation of 30 points.

Task: Test the school district's claim at a 5% significance level.

Question 8: Average Time Spent Exercising

Scenario: A fitness app claims that its users spend **at least** 45 minutes exercising daily. A sample of 95 users shows an average exercise time of 44 minutes with a population standard deviation of 4 minutes.

Task: Test the app's claim at a 5% significance level.

Question 9: Average Monthly Rent

Scenario: A real estate company claims that the average monthly rent in a particular city is \$1,500. A sample of 70 apartments shows an average rent of \$1,470 with a population standard deviation of \$80.

Task: Test the real estate company's claim at a 5% significance level.

Question 10: Average Weight Loss

Scenario: A weight loss program claims that participants lose **at least** 12 kg in 3 months. A sample of 50 participants shows an average weight loss of 11.5 kg with a population standard deviation of 1.8 kg.

Task: Test the program's claim at a 5% significance level

Submission

Please submit your Python code and results for each question in a Jupyter notebook. Make sure to include your hypothesis, Z-score, p-value, and conclusion for each question.
